

1 Explainability in Reinforcement Learning

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Chapter 1

Experiments

This chapter presents the experimental study of the Clustering Agent Representation (CAR) method introduced in the previous chapter. It focuses on the experimental validation of the method and its ability to consistently identify meaningful conceptual structures. The remainder of this chapter is organized as follows.

- Section 1.1 describes the software implementation of the CAR pipeline.
- Section 1.2 details the experimental configuration, including the environments, the reinforcement learning setup, and the training of the surrogate policies.
- Section 1.3 presents and analyses the results obtained by applying CAR and its metrics.
- Section ?? summarizes the findings and discusses their implications.

1.1 Software Implementation

Deep Learning (DL) and Reinforcement Learning (RL) are inherently empirical fields, in which the effectiveness of a method is assessed through experimentation. For this reason, particular attention is given to the implementation of CAR. It relies on established libraries that provide efficient and widely used implementations of its main components. The implementation is made publicly available as open-source to support the objective and reproducible replication of the reported results.¹

CAR is organized as a pipeline of five stages, shown in Figure 1.1. Each stage is an independent executable, and stages communicate only through artifacts written to disk. This decoupling allows any stage to be rerun or inspected independently. The parameters of each experiment are gathered in a single configuration object, described in Section 1.2. Every result can therefore be traced back to the settings that produced it.

Table 1.1 summarizes the main third-party libraries used at each stage of the pipeline.

¹https://github.com/Maxalaar/Clustering_Agent_Latent_Space

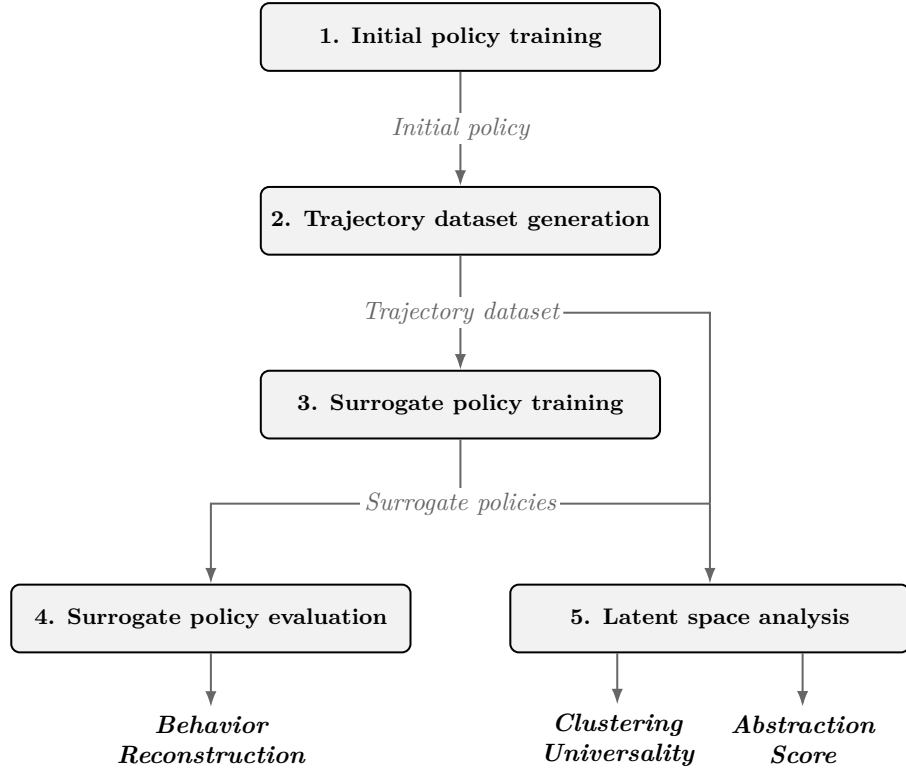


Figure 1.1: The CAR pipeline.

1.1.1 Initial policy training

The first stage trains the initial policy using RLlib [Moritz et al., 2018, Liang et al., 2018]. RLlib provides a highly scalable implementation of RL algorithms and supports a wide range of state-of-the-art methods. It supports distributed training, allowing the computational workload to be spread across multiple processes and machines.

The use of RLlib allows CAR to be applied to any environment implemented using the Gymnasium interface [Towers et al., 2024]. This provides a common interface for environments with different observation and action spaces. This flexibility allows CAR to be applied to a wide range of environments.

By default, the RL algorithm used for the initial policy is Proximal Policy Optimization (PPO) [Schulman et al., 2017]. Other algorithms supported by RLlib, such as Asynchronous Proximal Policy Optimization (APPO) [Espeholt et al., 2018], Deep Q-Networks (DQN) [Mnih et al., 2013], and DreamerV3, can also be selected through the same configuration.

The policy and critic networks are implemented using PyTorch [Paszke et al., 2019]. PyTorch provides efficient GPU computation and supports a wide range of neural network architectures. This provides a standardized implementation that is compatible with the broader DL ecosystem.

The output of this stage is a checkpoint containing the trained policy.

Pipeline Stage	Library	Description
Initial policy training	RLlib [Moritz et al., 2018, Liang et al., 2018]	Distributed RL library used to train the initial policy.
	PyTorch [Paszke et al., 2019]	Deep learning framework implementing the policy and critic networks.
	Gymnasium [Towers et al., 2024]	Common interface used to plug in any RL environment.
Trajectory dataset generation	RLlib [Moritz et al., 2018, Liang et al., 2018]	Distributed environment runners used to collect trajectories in parallel.
	Apache Arrow [Apache Arrow, 2016]	Columnar file format (Parquet) used to store the trajectory datasets.
Surrogate policy training	PyTorch Lightning [Falcon, 2019]	Structured training loop used for the behavior cloning of the surrogate policies.
Surrogate policy evaluation	RLlib [Moritz et al., 2018, Liang et al., 2018]	Wraps each surrogate policy as an RL policy to evaluate it in the environment.
Latent space analysis	cuML [RAPIDS, 2023]	GPU-accelerated clustering algorithms used to analyze the latent representations.

Table 1.1: Main software libraries used at each stage of the CAR pipeline.

1.1.2 Trajectory dataset generation

The second stage builds the dataset used to train the surrogate policies. The distributed execution capabilities of RLlib are used to collect trajectories in parallel. Multiple environment runners execute the initial policy simultaneously, distributing the computational workload across the available resources.

The initial policy is run in the environment, and the interactions between the agent and the environment are recorded. For each step, the dataset stores the observation together with the parameters of the action distribution produced by the policy. Trajectories are stored as Parquet files [Apache Arrow, 2016]. Parquet is a compressed columnar format widely used for large-scale data processing. Its columnar structure allows data to be read efficiently and supports concurrent access from multiple processes. This is particularly useful here because the datasets contain on the order of one million observation–action pairs and cannot be loaded entirely into memory.

1.1.3 Surrogate policy training

The third stage trains the surrogate policies using behavior cloning. This stage therefore converts the original RL problem into a supervised learning problem. The surrogate policies are trained to reproduce the action distributions generated by the initial policy .

The training procedure is implemented using PyTorch Lightning [Falcon, 2019]. PyTorch Lightning provides a structured interface for supervised training while remaining compatible with the PyTorch ecosystem. It handles common training operations such as data loading, optimization, checkpointing, and GPU execution, while allowing the underlying neural network implementation to remain fully controlled.

Once training is complete, the stage produces a set of multiple trained

79 surrogate policies.

80 1.1.4 Surrogate policy evaluation

81 The fourth stage measures how faithfully each surrogate policy reproduces the
 82 behavior of the initial policy. Each surrogate is wrapped so that it can be used
 83 as an RLlib policy and is then evaluated in the original environment. Its mean
 84 cumulative reward over multiple episodes is compared with that of initial policy.
 85 Evaluation episodes are collected using parallel environment runners.

86 This stage provides the value of the *Behavior Reconstruction*. It therefore
 87 verifies whether the surrogate policies adequately reproduce the behavior of the
 88 initial policy before their latent representations are analyzed. If a surrogate does
 89 not reproduce the original behavior sufficiently well, its training can be repeated
 90 before proceeding to the next stage.

91 1.1.5 Latent space analysis

92 The fifth stage performs the main analysis of CAR. The latent representations
 93 produced by the surrogate policies form the point clouds that are analyzed using
 94 clustering algorithms.

95 The clustering step relies on the cuML library [RAPIDS, 2023], which pro-
 96 vides GPU-accelerated implementations of several clustering algorithms. This
 97 significantly reduces the computational cost of clustering and makes it possible
 98 to analyze point clouds containing tens of thousands of representations.

99 The resulting partitions are used to compute the *Clustering Universality* and
 100 the *Abstraction Score*. The outputs of this stage are the resulting metric values
 101 and the material required for the qualitative analysis presented in the following
 102 sections.

103 1.2 Experimental Configuration

104 This section describes the configuration of the experiments reported in this
 105 chapter. It covers the environments, the neural architecture of the policies, and
 106 the main parameters of each pipeline stage. In general, for each environment, a
 107 single initial policy is trained and then imitated by three surrogate policies.

108 1.2.1 Environments

109 All experiments use environments from the Atari benchmark [Bellemare et al., 2013],
 110 accessed through the Arcade Learning Environment. The use of Atari environ-
 111 ments offers several advantages. First, Atari is a widely used benchmark in
 112 the RL community, with many studies using these environments as standard
 113 evaluation settings. This makes the experimental results easier to compare
 114 with existing work. Second, these environments were not specifically designed
 115 for RL, which reduces the risk of introducing biases related to the design of
 116 the environments themselves. Finally, the benchmark provides a diverse set of
 117 well-known environments with varying levels of complexity. This diversity makes
 118 it possible to evaluate CAR across different types of tasks and levels of difficulty.

The configuration of the Atari environments follows common practice in RL research [Machado et al., 2017].

In each environment, the agent receives visual observations in the form of raw pixels. Each image is resized to 64×64 pixels and converted to grayscale to reduce memory requirements. An exception is made for Pac-Man, where color information is important for determining the behavior of other agents in the environment. The original color information is therefore preserved for this environment.

To provide information about the dynamics of the environment, four consecutive observations are stacked along the channel dimension. The resulting observation has a shape of $64 \times 64 \times 4$. This allows the agent to infer the motion of objects from successive observations rather than relying on a single frame.

Frame skipping is also used to reduce the duration of each episode. The agent selects a new action every five frames, while the selected action is repeated for the intervening frames. This provides sufficient temporal resolution for effective control while reducing the number of decisions required during an episode.

The actions correspond to the different buttons available on the Atari controller. The resulting action space is therefore discrete. Since Atari games are fully deterministic, sticky actions are introduced to add stochasticity to the trajectories. With a probability of 0.05, the environment repeats the action selected at the previous step instead of applying the new action. This prevents the agent from simply memorizing a fixed sequence of actions during training.

With the environment configuration established, the next section describes the objective associated with each environment. A visualization of the six environments is provided in Figure 1.2 to give an overview of the experimental settings.

Breakout. The agent moves a paddle at the bottom of the screen (Figure 1.2a) to bounce a ball into a wall of bricks, each broken brick scoring a point, and the ball speeding up as the wall is cleared.

Mario Bros. A single-screen platform game (Figure 1.2b). The player clears each phase by hitting enemies from below to flip them, then kicking them off the platforms before they recover. A central POW block flips every enemy at once but can only be used a few times per phase.

Ms. Pac-Man. The player moves through a maze (Figure 1.2c), eating pellets while avoiding four ghosts, with power pellets briefly making the ghosts edible. The maze structure and the many possible local situations, such as a nearby ghost, an intersection, or an active power pellet, produce a large number of distinguishable states.

Pong. A minimal two-player game (Figure 1.2d) in which the agent controls one paddle and must send the ball past the opponent.

Space Invaders. The agent moves a cannon along the bottom of the screen (Figure 1.2e), firing at descending rows of aliens while dodging their shots and using destructible shields, with a bonus saucer occasionally crossing the top.

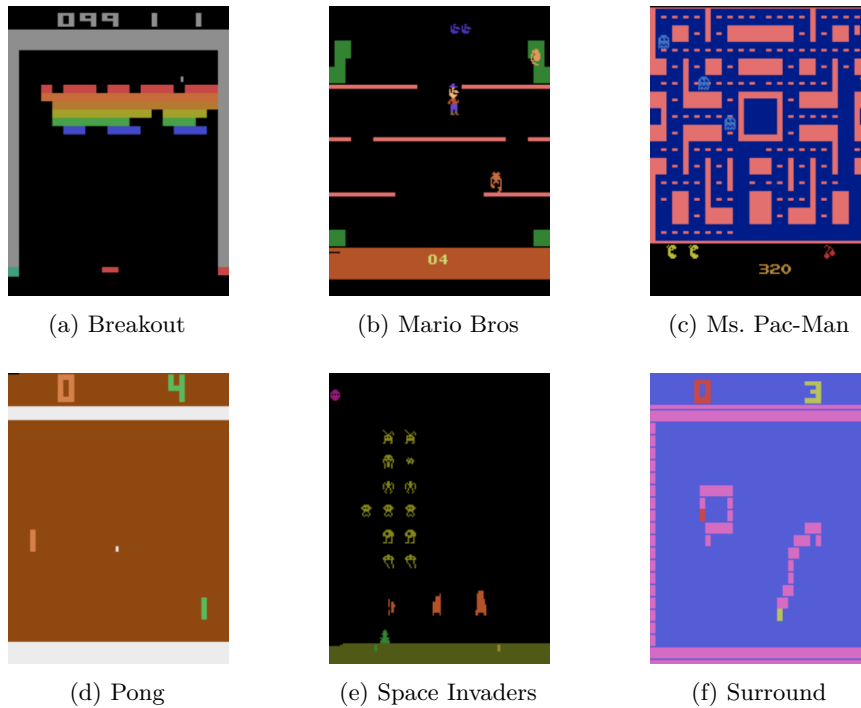


Figure 1.2: The six Atari environments used to evaluate CAR.

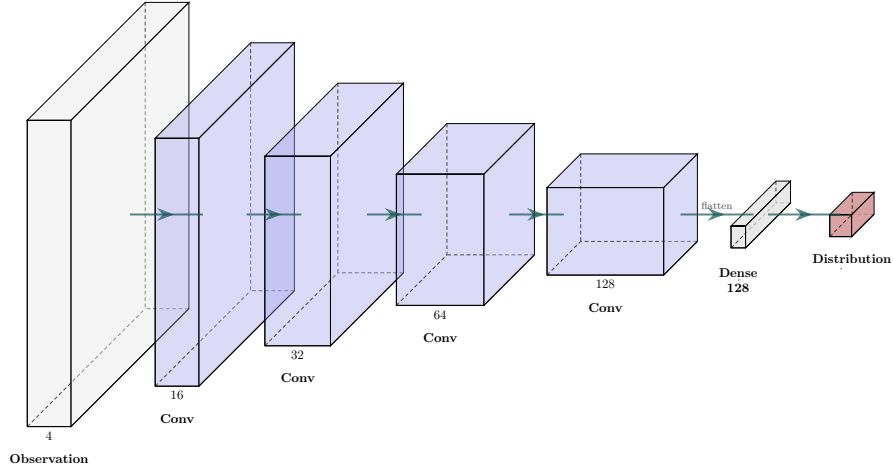
Surround. A two-player game on a grid (Figure 1.2f). Each player leaves a solid trail behind a moving cursor, and the goal is to force the opponent to crash into a trail or a wall.

1.2.2 Neural architecture

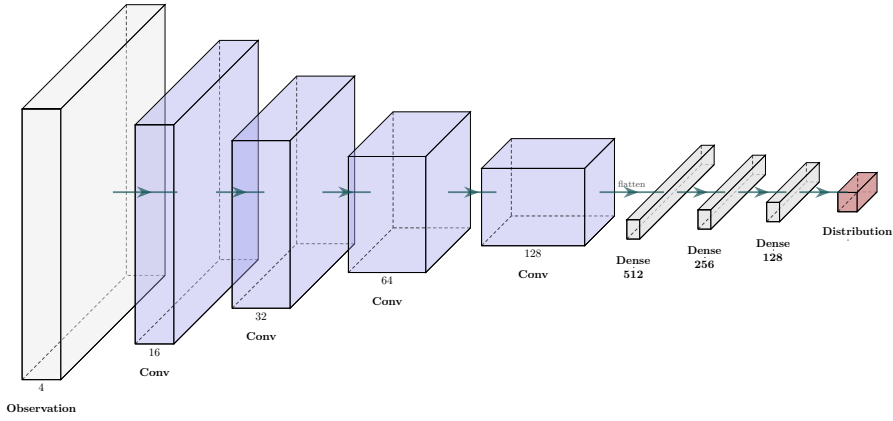
The surrogate policies use a convolutional neural network to process the visual observations. The input consists of a stack of four consecutive 64×64 frames. The convolutional part consists of four layers with 16, 32, 64, and 128 channels, using 4×4 kernels with a stride of 2. Each convolutional layer is followed by a LeakyReLU activation. The resulting representation is then flattened and passed through a multilayer perceptron that produces the parameters of the action distribution.

The architecture of the multilayer perceptron varies across environments. For Pong and Space Invaders, it contains a single hidden layer with 128 units. For the other four environments, it contains three hidden layers with 512, 256, and 128 units. Figure 1.3 illustrates both configurations. The size of the output layer matches the number of actions available in each environment: 4 for Breakout, 18 for Mario Bros, 9 for Ms. Pac-Man, 6 for Pong, 6 for Space Invaders, and 5 for Surround.

The latent space analyzed by CAR is not restricted to a specific layer or architecture. Any latent space produced by the network can be analyzed using the same methodology. In the experiments presented in this chapter, the default analysis considers the representation produced by the last convolutional layer.



(a) Pong and Space Invaders.



(b) Breakout, Mario Bros, Ms. Pac-Man, and Surround.

Figure 1.3: Architecture of the surrogate policies.

184 The layer-wise analysis additionally considers the representations produced by
 185 the hidden layers of the multilayer perceptron. The final output layer is excluded
 186 because it directly produces the action-distribution parameters.

187 1.2.3 Parameters

188 Having defined the environments and neural architectures, the remaining pa-
 189 rameters of the CAR pipeline must now be specified. There are too many
 190 parameters to discuss them all in detail, as each stage of the pipeline introduces
 191 its own configuration choices. This is partly due to the combination of three
 192 distinct fields, deep learning, reinforcement learning, and clustering. The main
 193 parameters shared across the experiments are summarized in Table 1.2.

194 Given the large number of possible configurations, it is not feasible to evaluate

Environment configuration	
Frame size	64×64
Frames stacked per observation	4
Frame skip	5
Sticky-action probability	0.05
Initial policy training	
Algorithm	PPO
Optimizer	Adam
Learning rate	1.5×10^{-4}
GAE λ	0.95
Policy clipping ε	0.1
Value-function clipping	10
Entropy coefficient	0.01
KL coefficient	0.5
Epochs per update	10
Gradient clipping (global norm)	100
Surrogate policy training	
Surrogate policies per environment	3
Observation–action pairs per dataset	$\geq 10^6$
Batch size	2048
Optimizer	Adam
Learning rate	1×10^{-4}
Latent space analysis	
Embedding per surrogate policy	50 000
Clustering algorithm	k -means
Distance	Euclidean distance
Number of clusters K	2 to 30

Table 1.2: Main experimental parameters.

all possible combinations. The choices made in this work are therefore based primarily on standard practices from the literature and on practical experience. Whenever possible, simple and widely used configurations are preferred. For the clustering algorithm and distance metric in particular, the simplest suitable choices are used to establish a baseline.

These choices do not restrict the CAR methodology itself. Most components can be replaced or configured differently depending on the application. The evaluation metrics introduced in the previous section provide an objective basis for comparing these alternatives and determining which configurations produce the most consistent and meaningful results.

1.3 Results

This section reports the results of applying CAR to the six environments described in Section 1.2. Sections 1.3.1 and 1.3.2 first check that the two learning stages of

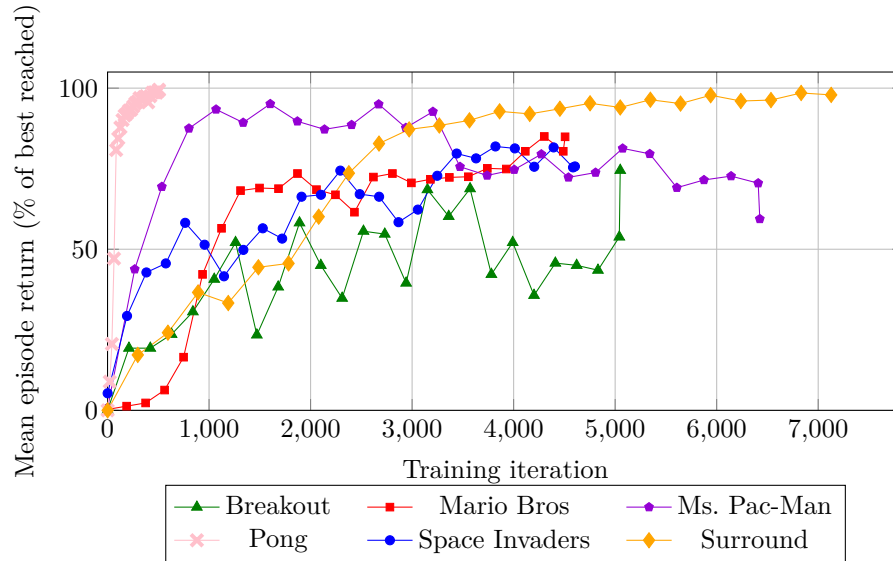


Figure 1.4: Training of the initial policies with PPO, one curve per environment.

the pipeline behave as expected: that the initial policies reach a competent level of play, and that the surrogate policies faithfully imitate them. Section 1.3.3 then presents the analysis of the latent representations and the values of the three evaluation metrics.

1.3.1 Initial policy training

Figure 1.4 shows how the initial policies improve during PPO training. For each environment, the mean episode return is normalized so that 0% and 100% correspond to the lowest and highest mean return observed during that run. In every environment the return increases and then reaches a plateau, at which point training is stopped. The number of iterations required to reach this plateau varies from a few hundred, for Pong, to several thousand, for Surround and Ms. Pac-Man. For Ms. Pac-Man the return decreases after its peak; since the best checkpoint is retained rather than the last one, this does not affect the policy that is subsequently analyzed.

To confirm that the trained policies play competently, their scores are compared with the reference scores reported by [Machado et al., 2017]. Each initial policy reaches a score of the same order as these references, so it constitutes a reasonable target for imitation.

1.3.2 Surrogate policy training

Figure 1.5 shows, for each environment, the training action loss of the three surrogate policies: the mean squared error between the action-distribution parameters of the surrogate and those of the initial policy. In every case the loss decreases smoothly, and the three independently trained policies follow almost identical trajectories, converging to the same value; the three curves of each environment largely overlap. This consistency matters for CAR: the method

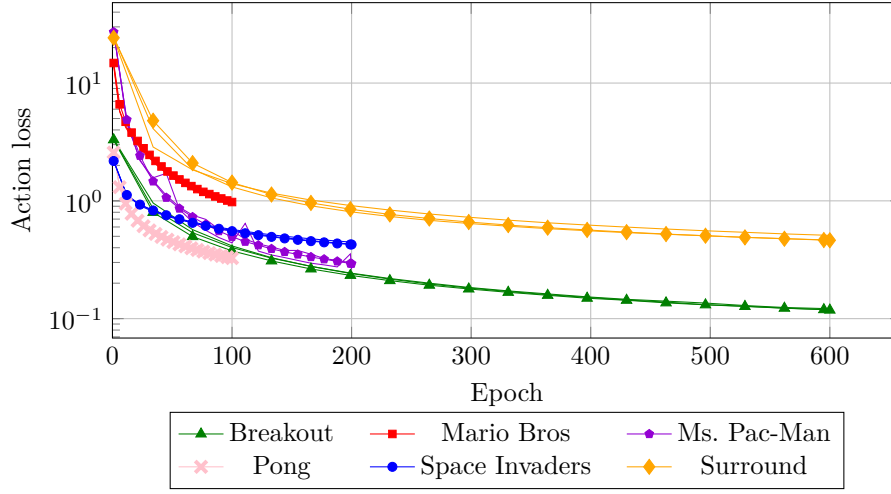


Figure 1.5: Training action loss of the surrogate policies, on a logarithmic scale, three curves per environment.

233 compares the latent representations of independently trained policies, and this
 234 comparison is only meaningful if these policies implement the same decision
 235 function. The behavioral fidelity of the surrogate policies is quantified by the
 236 *Behavior Reconstruction* in the next section.

237 1.3.3 Latent space analysis

238 We first evaluate the ability of the surrogate policies to replicate the behavior of
 239 the initial policies (*Behavior Reconstruction*, Table 1.3). The surrogate policies
 240 reach a *Behavior Reconstruction* of 0.96 on average over 100 evaluation episodes,
 241 ranging from 0.83 for Breakout to 1.00 for Ms. Pac-Man and Surround. They
 242 therefore reproduce the decisions of the initial policies closely enough for their
 243 latent representations to be analyzed in place of those of the initial policies.

Environment	<i>Behavior Reconstruction</i>	<i>Abstraction Score</i>	<i>Clustering Universality</i>
Breakout	0.83 $\pm$ 0.03	0.82 $\pm$ 0.07	0.48 $\pm$ 0.11
Mario Bros	0.99 $\pm$ 0.03	0.71 $\pm$ 0.14	0.55 $\pm$ 0.10
Ms. Pac-Man	1.00 $\pm$ 0.03	0.43 $\pm$ 0.09	0.65 $\pm$ 0.08
Pong	0.95 $\pm$ 0.01	0.99 $\pm$ 0.00	0.49 $\pm$ 0.05
Space Invaders	0.97 $\pm$ 0.07	0.69 $\pm$ 0.06	0.67 $\pm$ 0.07
Surround	1.00 $\pm$ 0.01	0.55 $\pm$ 0.07	0.61 $\pm$ 0.09

Table 1.3: Evaluation of *Behavior Reconstruction*, *Abstraction Score* and *Clustering Universality* across the six Atari environments.

For the remainder of the analysis, a random subset of 50,000 observations is embedded into the Latent Space (LS) of each surrogate policy, using the representation extracted directly after the last convolutional layer, and the resulting point cloud $\mathcal{E}$ is partitioned with k -means clustering [MacQueen, 1967].

Figure 1.6 reports the evolution of *Clustering Universality* with respect to the number of clusters. After the first few values of K , every curve settles onto a plateau whose level depends on the environment, from about 0.45–0.50 for Breakout and Pong to about 0.65 for Ms. Pac-Man and Space Invaders. Since *Clustering Universality* is computed from the Adjusted Rand Index (ARI), and an ARI of 0.5 already corresponds to substantial agreement [Amezquita et al., 2020], these values indicate that independently trained surrogate policies recover a similar cluster structure in every environment, and that this structure is stable with respect to the number of clusters. The strength of this agreement varies across environments, but remains clearly above chance everywhere. The absence of a single optimal number of clusters suggests that K should be chosen according to the level of granularity desired for the analysis.

The *Abstraction Score*, in contrast, varies markedly across environments. It is close to 1 for Pong (0.99) and high for Breakout, Mario Bros and Space Invaders (0.69 to 0.82): in these environments the latent clusters are largely independent of a direct clustering of the observations or of the action distributions. It is lower for Surround (0.55) and Ms. Pac-Man (0.43). In these two environments the position of the agent on the screen largely determines both the raw observation and the decision to take, so a clustering of the LS naturally coincides with a clustering of the observations, and the *Abstraction Score* cannot separate the two; a low score does not by itself imply that the representation is not abstract.

With fewer clusters, CAR highlights broad strategic patterns. For instance, in the Mario Bros environment, the two main groups correspond to whether the POW block is present (highlighted in red in Figure 1.8A) or not. This is particularly interesting, as the use of the POW block is a key determinant of strategy in this environment [AtariAge, 2024].

With a larger number of clusters, the analysis captures more fine-grained tactical behaviors, such as “Attacking the left” and “Navigating through the level” (see Figure 1.8B).

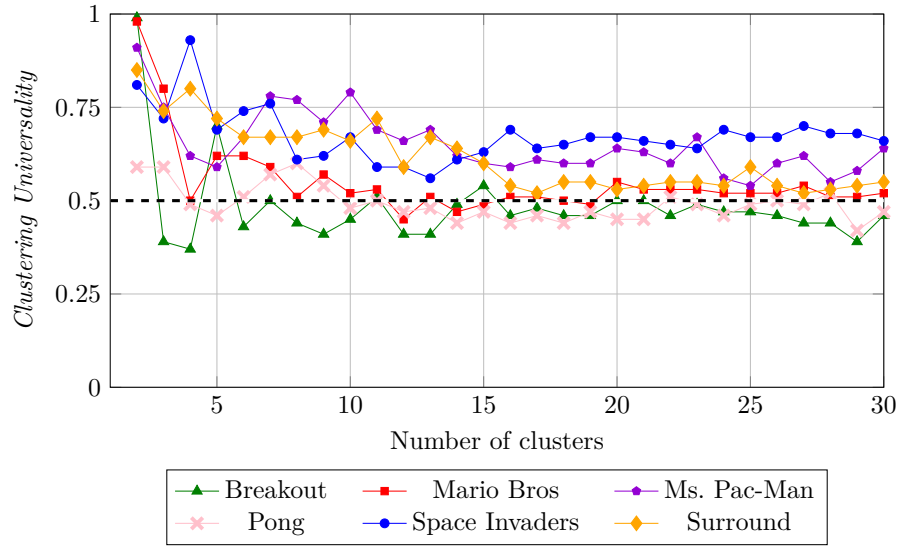


Figure 1.6: Evolution of *Clustering Universality* with respect to the number of clusters, for each of the six Atari environments.

Both perspectives are valid, depending on whether the goal is to study strategic or tactical behaviors. Similar interpretable structures can be observed across the evaluated environments.

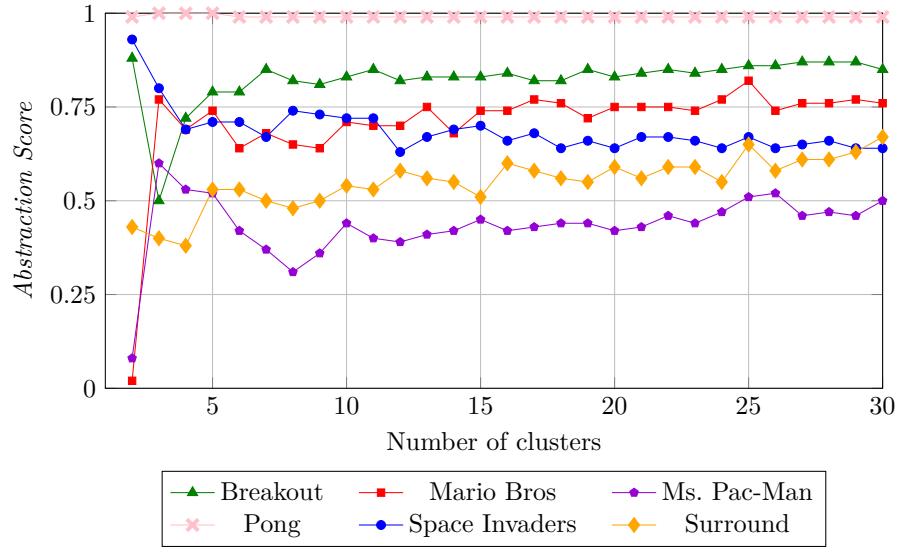
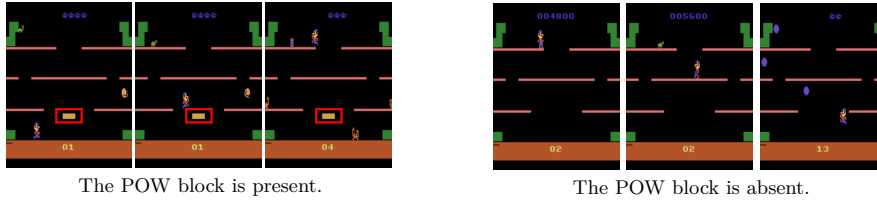


Figure 1.7: Evolution of *Abstraction Score* with respect to the number of clusters, for each of the six Atari environments.

(A) The case with 2 clusters.



(B) The case with 5 clusters.

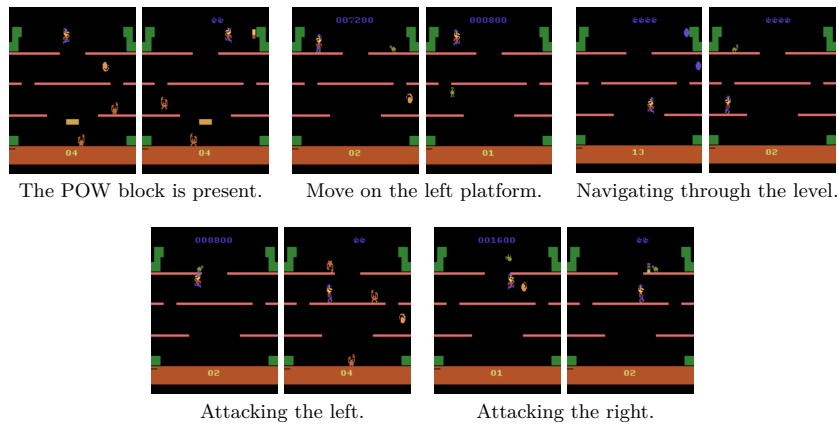


Figure 1.8: Effect of the number of clusters on the groupings identified by CAR in the Mario Bros environment.

Acronyms

APPO Asynchronous Proximal Policy Optimization 2

ARI Adjusted Rand Index 11

CAR Clustering Agent Representation 1–4, 6–9, 11, 13

DL Deep Learning 1, 2

DQN Deep Q-Networks 2

GPU Graphics Processing Unit 3

LS Latent Space 11

PPO Proximal Policy Optimization 2, 8, 9

RL Reinforcement Learning 1–5

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